LP2 -P6(Expert System)

6: Implement Expert System for Hospitals and medical facilities.

```
def check(msg):
    ans = input(msg).lower()
    if ans not in ['y', 'n']:
        print("Error: Enter only y or n")
        return None
    return ans


def expert_system():
    print("---- Hospital Expert System ----")
    name = input("Enter Patient Name: ")
    age = int(input("Enter Age: "))
    print("Select Symptoms (y/n):")
    fever = check("Fever? : ")
    if fever is None:
        return
    cough = check("Cough? : ")
    if cough is None:
        return
    chest = check("Chest Pain? : ")
    if chest is None:
        return
    stomach = check("Stomach Pain? : ")
    if stomach is None:
        return
```

```
    print("\n---Analyzing Symptoms ---")

    if chest == 'y':
        Department = "Cardiology"
        Advice = "Consult heart specialist immediately."

    elif fever == 'y' and cough == 'y':
        Department =  "General Medicine"
        Advice = "Possible infection or viral fever."

    elif stomach == 'y':
        Department = "Gastroenterology"
        Advice = "Consult stomach specialist."

    else:
        Department = "Outpatient (OPD)"
        Advice = "No critical symptoms. You may proceed to OPD."

    print(f"\n=== Patient Report ===")
    print(f"Name: {name}")
    print(f"Age: {age}")

    print(f"Recommended Department: {Department}")
    print(f"Advice: {Advice}")

expert_system()
```

Viva Questions & Answers — Hospital Expert System

1. What is an Expert System?

An Expert System is an AI-based program that makes decisions using predefined rules and knowledge similar to a human expert.

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2. What type of expert system is your program?

Rule-based expert system.

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3. Why is it called rule-based?

Because decisions are made using if-elif-else rules.

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4. What is the purpose of your project?

To help patients identify the correct hospital department based on symptoms.

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5. Which language is used?

Python.

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6. What are the main concepts used?

- Functions
- Conditions
- User input
- Decision making
- Validation

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7. Why did you use functions?

To reduce repetition and improve code readability.

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8. What is the purpose of check() function?

It validates user input and ensures only y or n is accepted.

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9. What happens if invalid input is entered?

Program displays:

Error: Enter only y or n

and stops execution.

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10. Why did you use return None?

To indicate invalid input and stop further processing.

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11. What is the use of lower()?

To convert input into lowercase for easy comparison.

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12. Why did you use if fever is None?

To check whether validation failed.

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13. What is the role of if-elif-else?

It performs decision making based on symptoms.

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14. How does your expert system work?

It takes symptoms as input, analyzes them using rules, and recommends department and advice.

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15. Which departments are included?

- Cardiology
- General Medicine
- Gastroenterology
- OPD

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16. Why is chest pain checked first?

Because it may indicate a serious heart problem requiring urgent attention.

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17. What is OPD?

Outpatient Department.

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18. Can your system replace doctors?

No, it only provides basic guidance.

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19. What are limitations of your expert system?

- Uses fixed rules
- Cannot learn automatically
- Limited symptoms

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20. How can you improve this system?

- Add database
- Add more diseases
- Use Machine Learning
- Use NLP
- Add GUI

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21. Difference between Expert System and Chatbot?

Expert System	Chatbot
Gives expert decisions	Gives conversational replies
Uses knowledge rules	Uses interaction patterns
Problem solving	Communication

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22. What is Artificial Intelligence?

AI is the simulation of human intelligence by machines.

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23. Is your program AI?

Yes, it is a simple AI rule-based expert system.

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24. Why is validation important?

To prevent incorrect input and improve reliability.

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25. What is the advantage of expert systems?

- Fast decision making
- Reduces human effort
- Available anytime

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26. What are symptoms in your system?

Inputs used to identify possible medical condition.

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27. Why did you choose hospital expert system?

Because it is useful for basic patient guidance and department recommendation.

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28. What will happen if multiple symptoms are true?

The first matching condition executes.

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29. What is the use of input()?

To take user input from keyboard.

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30. What is the use of print()?

To display output to the user.